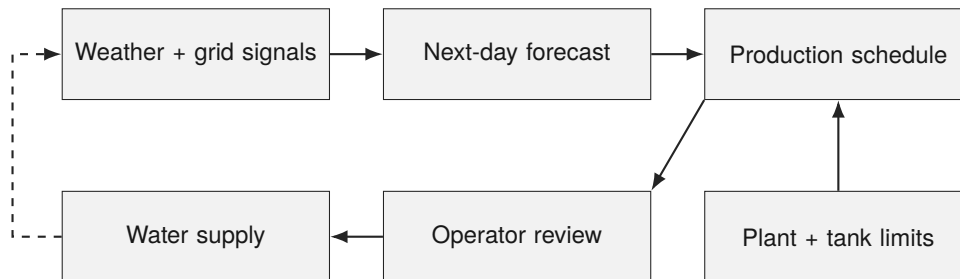


01 / SOLUTION DESCRIPTION

Schedule around storage.

Use electricity that would otherwise be curtailed to produce drinking water. Store water for evening demand. This requires spare production capacity, usable tank storage and coordination with the grid operator.



Proposed pilot loop. Planning uses weather; grid reports are retrospective context.

Current implementation

Historical Paphos weather feeds a LightGBM radiation reference. A constrained scheduler plans one unit and one tank. The workspace connects assets, plans, evaluation, grid evidence and review records. Pilot inputs checks declared observation files, units, gaps and first-available times separately from planning. Streamlit retains the local experiment. No controller or live plant is connected.

Plant scenario	Fixed setting
Production limit	500 m ³ /h
Hourly demand	120 m ³ /h at the default temperature
Tank	4,000 m ³ ; safety minimum 20%
Start / end storage	50% / 50%
Specific electricity	3.4 kWh/m ³
Assumed tariff	€101/MWh midday; €183 evening; €130 otherwise

The water balance

$$S_{h+1} = S_h + q_h - d_h, \quad S_{\min} \leq S_h \leq S_{\max}, \quad 0 \leq q_h \leq q_{\max}.$$

S is stored water, q production and d demand. The current solver minimises the assumed tariff cost; forecast sunshine breaks ties between equally priced hours. Both schedules finish with the same water and storage. Ramp rates, maintenance and minimum run times need operator data.

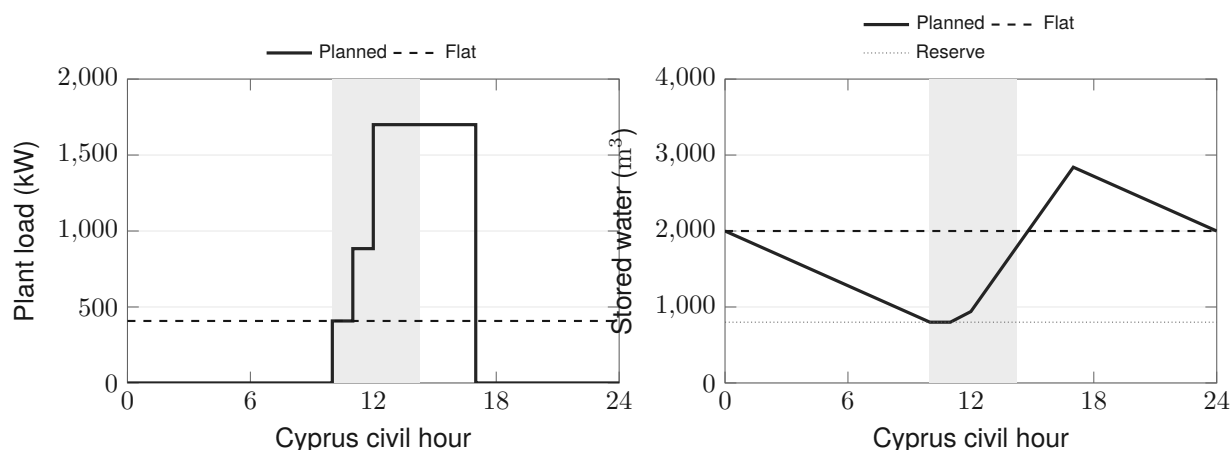
Why test machine learning

Weather changes the useful production windows. Machine learning combines weather and calendar signals to test whether that timing can be anticipated. Superiority is unproven: the current reference loses on MAE and adds no tariff saving over price-only scheduling. Retain AI only if a new pilot beats competent controls on matched water service, cost or measured renewable absorption. Sunshine itself is not a curtailment measurement.

02 / EVIDENCE

Inspect one day.

15 July 2026 / 4,000 m³ tank. The existing scenario moves production into midday and draws stored water later. The shaded band is EAC's reported Group 1 curtailment window, 10:00–14:15.



Scenario output; reported window interpreted as Cyprus civil time. EAC July report, page 15.

During that window, planned load is 5,117 kWh; flat load is 1,734 kWh. The price-only control also gives 5,117 kWh. This is timing overlap, not recovered solar energy. The grid report never enters the scheduler. Across 62 reported days and five tank sizes, 310 matched cases were checked; September coverage is unknown.

Forecast control

The independent reference is not Stefanos's model. Training ends before July; July–September contains 2,208 hours. MAE: 8.812 model / 7.638 safe persistence. RMSE: 17.942 / 19.433 W/m². Issue time is previous-day 18:00; persistence uses yesterday through hour 18 and two days earlier after that. The 600 W/m² threshold was selected in June.

Matched scheduling result

Default fixture	Flat schedule	Planned schedule
Water produced (m ³)	2,880	2,880
Electricity (kWh)	9,792	9,792
Scenario cost (€)	1,319.88	988.99
Tank safety violations	0	0

The lower cost comes from assumed clock prices. Across 92 matched days, the model added €0 tariff saving over a price-only schedule. Equal energy at a constant carbon factor saves zero carbon.

Checks and limits

Reconstruction weather is not sensor truth or an archived issued forecast. EAC's Group 1 fields are not national totals. Published percentages are retained; 11 daily reports have arithmetic discrepancies beyond displayed rounding. The synthetic garden and leak checks are demonstrations, not field validation.

03 / HANDOFF

Review the handoff.

Files and roles

Stefanos	Primary model, plant/tank assumptions and scheduler, following his roadmap. Supply weather, predictions, model metadata and schedule CSVs.
Loucas Louka	Original concept and philosophy; evaluation, dashboard, integration and prediction experiments. Recheck the incoming model against the same safe controls.
Andreas Nikolaides / Cleopas Cleopa	Business case, customer evidence and pitch. Technical review stays with Loucas and Stefanos.

Run and inspect

Open the online operations workspace, or Start `AquaShift.command` for the same interface locally. Change the date and tank size, inspect production and storage, compare forecasts and record a review. The package retains code, input data, model text, results and editable Blender scenes.

```
python -m unittest model.test_scheduler model.test_sites
python -m unittest eval.test_evaluation
python results/independent-review.py
```

Use the tested environment in `README.md`. Evaluate Stefanos's aligned hourly CSV in a separate result directory; retain both models' provenance.

Next evidence

Obtain one operator's capacity, storage, demand and constraints. Connect available curtailment signals and billed tariffs. Save forecasts before a new test period, then compare equal water service against current operation and a competent simple scheduler. The June residual experiment remains exploratory.

Submission review

The dashboard, video and documents are starting points. Confirm names, contact details, declarations and signatures; understand the results and rehearse before submission. No teammate's completed model or unaided coding is claimed.

Sources: Stefanos's 30 September AquaShift pitch and work split.

Evidence: `results/metrics.json`, `curtailment-overlap.json`, `curtailment-independent-review.json`.

Grid context: EAC distribution curtailment reports; dated page references and source hashes retained in the package.

Open-Meteo / ECMWF IFS supplies the weather. Adopted software: LightGBM, SciPy/HiGHS, scikit-learn, Streamlit and Plotly. Codex assisted code, tests, graphics and draft materials. Concept attribution follows Loucas's clarification and is not a claim of exclusive invention of solar desalination.